Subjectivity-Intersection Mathematics

Constitutive Closure and Stabilized O3 Return across Atomic, Molecular, and Cellular Systems

Formal Criteria, Failure-Guided Cross-Domain Tests, and Domain Boundaries

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Document status: Public Working Preprint, Version 4.2 Draft
Date: 13 August 2026
DOI: 10.5281/zenodo.21909933
Preceding public version DOI: 10.5281/zenodo.21877695
Repository: SIEL-Research/Subjectivity-Intersection-Mathematics

Abstract

Can a scientific object be defined not as a completed thing that subsequently has relations, but as a whole constituted by the generation and return of a relation among differentiated perspectives? This paper develops *Subjectivity-Intersection Mathematics* (SIM) as a staged preregistered program for making that explanatory reversal operational. A domain-relative operational perspective comprises a local carrier together with the distinctions, admissible transformations, and causal readouts available from its relational position; it is neither a synonym for a component nor a claim of consciousness. Perspective intersection can generate a domain-relative third term, O_3, that is reducible to no isolated perspective and re-enters the conditions of continued formation.

The operational foundation is a staged recurrent-perspective sequence in which a learned joint contribution emerged without an installed C state, O3 target, pair identifier, or carrier-specific loss and vanished in an equally competent disconnected additive control. The preserved 003R–006A record, including a preregistered failure, supports the grammar

$$(O_A, O_B) \xrightarrow{\text{reciprocal intersection}} O_3 \xrightarrow{\text{self-re-entry}} (O'_A, O'_B, O'_3).$$

The same grammar is tested on constitutive questions across domains using six operational confirmation criteria: generation, necessity, re-entry, specificity, registered empirical minimality, and representation covariance, with null discrimination throughout. Atomic, molecular, and cellular experiments respectively test finite-mass relational closure, a molecule-minus-isolated Hamiltonian carrier, and a distributed mode generated by boundary, metabolism, and information-repair dynamics. A within-bridge precursor, E011, established held-out representation covariance inside a frozen reduced cellular bridge but did not independently execute atomic, molecular, and cellular targets.

Two preregistered precursor tests rejected uniform absolute calibration while preserving a common causal ordering. E014 therefore tested the refined invariant directly across independent reduced atomic, molecular, and cellular engines without a shared scalar mediator: after a domain-native stabilization interval, correct O3 return must be closer to the registered domain state than continued removal or physically distinguishable mismatched return. This direction passed all atomic ($3/3$), molecular ($9/9$), and cellular ($8/8$) primary checkpoints. The evidence supports a cross-domain causal invariant of stabilized O3-return specificity over the tested realizations, not a common O3 substance, common metric, microscopic atom-to-cell law, phenomenal subjecthood, or living-cell confirmation.

Keywords: subjectivity-intersection mathematics; perspective intersection; third perspective; O3; constitutive closure; atom; molecule; cell; representation covariance; self-re-entry; preregistration

Principal contributions

1. The primary mathematical object is generalized from reciprocal perspective intersection to a domain-relative constitutive-closure structure with explicit references, interventions, re-entry, whole-formation criteria, and representation equivalence.
2. A common grammar maps differentiated operational perspectives to a generated third term and its return: $(O_1^D, \ldots, O_n^D) \to C_D \to (O_1^{D\prime}, \ldots, O_n^{D\prime}, C'_D)$.
3. The 003R–006A sequence supplies the operational foundation: emergence, non-reduction, joint nonseparability, pair specificity, bilateral action, transport, and self-re-entry.
4. Atomic evidence supplies a complete mixed record rather than success selection: supported transfer, a rejected postulate, a supported isotope-ratio generator, an unresolved precision-limited test, and an open prediction.
5. Molecular Experiment 010 tests constitution of a four-centre whole by a complete carrier across three basis families and representation changes.
6. Cellular Experiment 009 tests natural emergence and self-re-entry of a distributed third mode as the constitutive condition of reduced-model life closure.
7. Experiment 011 is retained as a held-out representation-covariance test inside a frozen reduced bridge, not an independent atom–molecule–cell test.

8. Experiments 012 and 013 preserve two preregistered failures: mismatch specificity was representation-sensitive, and a common absolute score threshold did not transfer.
9. Experiment 014 prospectively supports a narrower cross-domain invariant across independently executed targets: after domain-native stabilization, correct O3 return is closer to the registered state than removal or mismatched return.
10. A compression ladder limits inference from domain construction through intervention and readout to the shared sign invariant; scalar agreement is not inverted into common source identity.
11. All conclusions are indexed by evidence class and retain their negative, mixed, open, and model-limited boundaries.

Contents

1 Introduction: from relation to intersection

Interactions are commonly described through messages, shared variables, latent edges, coordination patterns, or models one participant holds of another. These descriptions begin with participants and characterize what passes between them. Subjectivity-Intersection Mathematics asks a different question: can the reciprocal taking-up of one perspective by another generate a perspective that belongs to neither perspective separately and then participate in the formation of both?

The distinction matters. A relation may exist whenever A affects B. A perspective intersection requires two differentiated perspectives and reciprocal transformation: A receives B through A's own state organization, while B receives A through B's own. The two directed transformations may remain separate. The stronger possibility is that they become jointly nonseparable and function as a third perspective.

Let O_A and O_B denote the operational perspectives of A and B. The canonical structure of the program is

$$(O_A, O_B) \xrightarrow{\mathcal{X}} O_3 \xrightarrow{\mathcal{E}} (O'_A, O'_B, O'_3), \tag{1}$$

where $\mathcal{X}$ is reciprocal intersection and $\mathcal{E}$ is self-re-entry through unchanged dynamics. The first arrow is not averaging, fusion, consensus, or the installation of a shared memory. The second is not an external feedback channel added after the fact. It states that the generated perspective becomes part of the conditions under which the next two perspectives and their next intersection are formed.

This paper reports a sequence of preregistered tests of that grammar. Early experiments constructed an explicit carrier to establish operational sufficiency. Later experiments removed the carrier from the architecture and trained only an ordinary reciprocal task. The 006-series asked whether the emergent distributed result was causal, temporally recurrent, architecture-general, pair-specific, and one jointly nonseparable contribution rather than two directed memories.

The new cross-domain sequence asks a different but connected question: can the same explanatory order define what makes an atom, molecule, or cell continue as one constituted system? The transfer does not reuse a neural carrier or declare every physical interaction to be O3. It requires a separate operational bridge in each domain, a frozen isolated reference, an intervention that can destroy or alter the proposed joint term, and a readout of return into the constituents. E011 first tests whether closure identity belongs to a raw coordinate or to a relation preserved under a registered change of representation. E012–E014 then move beyond that single bridge: independent atomic, molecular, and cellular engines are exposed to removal, correct return, and mismatched return so that the proposed cross-domain invariant can fail prospectively.

2 A mathematical definition of perspective intersection

2.1 Operational perspective

A perspective is not identified with an input token or stored fact. It is the state-dependent way in which a system receives, transforms, and continues from what is encountered. For participant $i \in \{A, B\}$, define

$$O_i(t) = \big(h_i(t), \mathcal{H}_i(t), F_i, D_i\big), \tag{2}$$

where h_i is the current persistent state, $\mathcal{H}_i$ its history, F_i its update organization, and D_i its output map. Two systems can receive the same symbol yet instantiate different perspectives because their histories and transformations differ.

In the reciprocal-recall experiments, discrete symbols are minimal probes, not perspectives themselves. The operational perspective is the recurrent organization through which A incorporates B-originating information and through which B incorporates A-originating information.

2.2 Reciprocal intersection

Let the coupled updates be

$$h_A(t+1) = F_A(h_A(t), h_B(t), x_A(t)), \tag{3}$$
$$h_B(t+1) = F_B(h_B(t), h_A(t), x_B(t)). \tag{4}$$

Reciprocal intersection occurs when both cross-dependencies are active and alter subsequent state formation. It preserves distinction: h_A and h_B remain separate persistent states. Intersection is therefore not merger but mutually conditioned difference.

The two directed traces may be written

$$R_t^{A\leftarrow B} = h_A^{AB}(t) - h_A^{A\text{-only}}(t), \tag{5}$$
$$R_t^{B\leftarrow A} = h_B^{AB}(t) - h_B^{B\text{-only}}(t). \tag{6}$$

These traces show that each perspective has taken up the other. They do not yet establish a third perspective. That requires evidence that the two directed realizations participate in one nonseparable operation.

2.3 The third perspective O3

Define the joint post-learning intersection term

$$C_t = H_t(ab) - H_t(a0) - H_t(0b) + H_t(00), \tag{7}$$

where all four passes use identical weights, matched inputs, times, initial conditions, and evaluation episodes. The term removes the two one-sided histories and restores their common baseline. It is zero for a purely additive pair of disconnected directed memories.

Within this paper, C is identified as an operational third perspective O_3 when the following conditions hold:

1. it is absent as a named state, target, label, or specialized loss before learning;
2. it appears only under reciprocal interaction and post-learning counterfactual extraction;
3. it is reducible to neither O_A nor O_B under the frozen decomposition;
4. its A-side and B-side realizations are jointly nonseparable against an equally competent additive control;
5. its selective deletion changes both receivers;
6. its replacement by an equal-norm component from another pair loses the original function;
7. it contributes through unchanged dynamics to the next participant states and next extracted intersection.

This definition does not require a central O3 register. O3 can be distributed across A and B while remaining one operation. Its unity is functional and counterfactual: the two locations cannot be separated without losing the measured joint effect.

2.4 Self-re-entry

Let $X_t = (h_A(t), h_B(t))$ and let $\mathcal{I}_{-C_t}$ remove only the extracted O3 contribution. Under unchanged update F,

$$X_{t+1} = F(X_t), \tag{8}$$

$$X_{t+1}^{(-C)} = F(\mathcal{I}_{-C_t}(X_t)), \tag{9}$$

$$\Delta X_{t+1}^{C} = X_{t+1} - X_{t+1}^{(-C)}. \tag{10}$$

Applying the frozen extractor $\mathcal{R}$ at the next state gives

$$\Delta C_{t+1}^{C} = \mathcal{R}(F(X_t)) - \mathcal{R}(F(\mathcal{I}_{-C_t}(X_t))). \tag{11}$$

Equation (10) tests return into A and B; Equation (11) tests return into the next intersection. Exact reconstruction compares the observed missing contribution with the contribution predicted through the unchanged update. Thus O3 is not merely retained information. It is a current constraint on what the next perspectives and next intersection can become.

3 General constitutive-closure grammar

3.1 Operational perspective is not a replaceable label

The two-participant construction can be generalized without assuming that all domains contain autonomous agents, memories, or phenomenal viewpoints. It cannot, however, be generalized by replacing perspectives with bare local terms. Such a replacement would erase the hypothesis under test and reduce the program to an ordinary decomposition of an interacting system.

For domain D, define a domain-relative operational perspective as

$$O_i^D = \big(L_i, S_i, T_i, Q_i\big), \tag{12}$$

where L_i is a differentiated local carrier, S_i is the state space distinguishable from its relational position, T_i is the class of transformations admissible at that position, and Q_i is the registered causal readout available from it. Thus L_i carries a perspective but is not identical to it. The word *perspective* marks the organized selectivity of distinctions, transformations, and consequences. It does not attribute consciousness or phenomenal experience to particles, molecules, cellular modules, or numerical variables.

Let

$$\mathbf{O}_D(t) = \big(O_1^D(t), \dots, O_n^D(t)\big). \tag{13}$$

A constitutive intersection map $\mathcal{X}_D$ and re-entry map $\mathcal{E}_D$ define

$$\mathbf{O}_D \xrightarrow{\mathcal{X}_D} C_D \xrightarrow{\mathcal{E}_D} \big(\mathbf{O}'_D, C'_D\big). \tag{14}$$

The transfer from subjectivity to physical and biological domains is therefore not a metaphor appended after the mathematics. It is the explicit bridge hypothesis: whether domain-relative standpoints intersect so that a generated relation participates in constituting the scientific object. If perspective is removed from Equation (14), the framework collapses into a generic coupling analysis and loses the constitutive claim that distinguishes SIM.

3.2 Constitutive-closure structure

For a registered domain bridge, define

$$\mathfrak{C}_D = \big(\mathbf{O}_D, \mathcal{R}_D, C_D, \mathcal{I}_D, \mathcal{E}_D, W_D, \sim_D\big), \tag{15}$$

where $\mathcal{R}_D$ constructs matched isolated or local-only references; C_D is the joint term generated relative to those references; $\mathcal{I}_D$ is a preregistered class of admissible interventions; $\mathcal{E}_D$ returns the candidate relation through unchanged or explicitly frozen dynamics; W_D is a vector of whole-formation readouts with a registered decision region Ω_D; and $\sim_D$ is an equivalence relation over admissible representations.

The use of a readout vector rather than an unrestricted binary label prevents the definition from installing the desired whole in advance. The experiment freezes measurable coordinates W_D, their acceptance region Ω_D, and relevant nulls before the confirmatory intervention. For state $z = (\mathbf{O}_D, C_D)$, write

$$\chi_D(z) = \mathbf{1}[W_D(z) \in \Omega_D]\,. \tag{16}$$

This indicator is a decision about the registered realization, not a universal definition of atom, molecule, or life.

3.3 Decision conditions

For this research program, the following conditions are proposed as operational confirmation criteria rather than universal metaphysical necessities. A candidate C_D counts as a confirmed constitutive closure only relative to the frozen bridge and evidence class when all applicable conditions pass:

1. **Generation.** $C_D = \mathcal{X}_D(\mathbf{O}_D; \mathcal{R}_D)$ is absent from the matched isolated construction and is not installed as a target, label, or unexplained primitive.

2. **Necessity.** For the registered destructive intervention $I^- \in \mathcal{I}_D$, the whole-formation decision or specified continuous readout is lost:

$$\chi_D(\mathbf{O}_D, C_D) = 1, \qquad \chi_D\big(I^-(\mathbf{O}_D, C_D)\big) = 0, \tag{17}$$

   or the preregistered effect threshold is crossed.

3. **Re-entry.** Returning the same candidate through the frozen relation and timing restores the registered readout:

$$\chi_D\big(\mathcal{E}_D(I^-(\mathbf{O}_D, C_D), C_D)\big) = 1. \tag{18}$$

4. **Specificity.** Equal-norm substitutes, cross-pair exchange, time shifts, incomplete sector returns, or other registered false returns do not reproduce the same restoration.

5. **Registered empirical minimality.** No tested proper subterm $c \subsetneq C_D$ reproduces the complete vector of necessity, bilateral or all-part return, reconstruction, and whole-formation results. This criterion applies only within the registered intervention class $\mathcal{I}_D$ and is not proof of absolute mathematical uniqueness.

6. **Representation covariance.** For every registered admissible representation map g, there exists a transformed realization $g(C_D)$ such that

$$\mathfrak{C}_D \sim_D g(\mathfrak{C}_D), \qquad \chi_D(z) = \chi_D^g(g(z)), \tag{19}$$

   while the intervention and re-entry diagram commutes to the registered tolerance.

Null discrimination is not a seventh decorative item but a requirement across all six conditions. Additive coupling, correlation, a latent mode, an order parameter, or an interaction Hamiltonian can be important without satisfying generation, selective loss, specific return, and covariance together. Conversely, not every domain experiment presently tests the maximal definition. Table 1 states the evidence level actually attained.

3.4 Cross-domain evidence gates

The structure in Equation (15) distinguishes constitutive closure from an arbitrary decomposition followed by intervention in four ways. References and admissible interventions are frozen; the proposed third term must be generated rather than merely named; destruction must be paired with specific return rather than effect alone; and identity is attached to a preserved intervention–re-entry structure rather than a preferred coordinate. Each experiment below freezes its own bridge, target, controls, and interpretation boundary. Negative, mixed, unresolved, and open outcomes remain part of the same record.

Within-domain closure confirmation and cross-domain transfer are distinct decisions. Let $X_D^0(t)$ denote the registered intact domain state, and let $X_D^+(t)$, $X_D^-(t)$, and $X_D^\times(t)$ denote states following correct O3 return, continued O3 removal, and physically distinguishable mismatched return. Each domain preregisters its own similarity or recovery readout S_D, stabilization time τ_D, and checkpoint set T_D. Define

$$\Delta_D^-(t) = S_D\big(X_D^+(t), X_D^0(t)\big) - S_D\big(X_D^-(t), X_D^0(t)\big)\,, \tag{20}$$
$$\Delta_D^\times(t) = S_D\big(X_D^+(t), X_D^0(t)\big) - S_D\big(X_D^\times(t), X_D^0(t)\big)\,. \tag{21}$$

The proposed stabilized O3-return signature is

$$\sigma_D(t) = \big(\operatorname{sgn}\Delta_D^-(t), \operatorname{sgn}\Delta_D^\times(t)\big) = (+,+), \qquad t \in T_D,\ t \geq \tau_D. \tag{22}$$

Cross-domain transfer of this signature requires Equation (22) to pass prospectively in every registered domain. The values of S_D, the state coordinates, the carrier, the stabilization interval, and the margin magnitudes need not be equal. What is compared is the preregistered causal direction after domain-native stabilization. This is an operational sufficient condition over the tested realizations, not a universal metaphysical necessity.

The inference boundary is represented by the compression ladder

$$\begin{aligned} &\text{source construction} \longrightarrow \text{O3 carrier or generator} \longrightarrow \text{intervention history} \\ &\qquad \longrightarrow \text{domain state} \longrightarrow \text{registered readout} \longrightarrow \sigma_D. \end{aligned} \tag{23}$$

Evidence normally travels from left to right. Agreement at a scalar readout or at σ_D does not license inversion to identity of source construction, physical substance, or microscopic mechanism. Conversely, prospective agreement of the causal direction across independently executed engines is stronger than numerical coincidence inside one shared scalar mediator.

3.5 Evidence classes

The word *evidence* covers several non-equivalent operations in this program. Table 1 prevents those operations from being silently merged.

Table 1: Evidence classes used in the cross-domain program.

Class	Directly tested	Claim boundary
Benchmark-grounded transfer	Frozen predictions compared with published atomic measurements	Leading registered observables, not replacement of full QED or universal atomic derivation
Standard-theory counterfactual	Frozen interventions inside finite-basis quantum chemistry	Constitutive standpoint inside standard theory, not a new force or exclusive numerical capacity
Reduced-model confirmation	Preregistered interventions and new stochastic seeds in cellular dynamics	Model closure, not laboratory confirmation in living cells
Reduced-bridge prediction	Held-out representations predicted by a frozen operational transformation	Representation covariance inside the bridge, not a microscopic natural cross-scale law
Independent cross-domain intervention	Separate domain engines, domain-native stabilization, removal, correct return, and physically distinguishable mismatch	Transfer of a registered causal direction, not equality of metrics, carriers, substances, or source mechanisms

4 Relation to neighboring concepts

4.1 Scientific objects, individuation, and process

SIM enters an established debate rather than an empty field. Process-oriented accounts challenge substance-first descriptions of organisms and other scientific objects, while work on individuation asks how a unity and its boundary arise rather than treating them as given [10, 11, 12]. Structural realism likewise assigns epistemic or ontological priority to preserved structure rather than intrinsic properties of isolated objects [13, 14]. SIM agrees that object identity cannot always be exhausted by a list of intrinsic component properties. Its additional burden is operational: it preregisters a domain bridge and tests whether a generated intersection term is necessary, specifically returnable, and covariant for a whole-formation readout.

The claim is therefore not that science has ignored relations, processes, or organization. It is that these can be placed in a prospective intervention–re-entry design whose explanandum is the constitution of the object rather than only its behavior after constitution.

4.2 Scientific representation, perspective, and invariance

Scientific representation has been analyzed through similarity, structural mappings, modeler purposes, and perspectival representation [15, 16, 17]. These traditions already show that a model does not transparently copy a target and that invariants can matter more than surface coordinates. SIM does not claim the discovery of invariance or perspective. It combines them in a stricter constitutive question: which representation changes preserve the registered generation, intervention, and re-entry identity of a third perspective?

Operational perspective in SIM is domain-relative but not merely observer-relative. Equation (12) binds a local carrier to its accessible distinctions, admissible transformations, and causal readouts. This makes perspective part of the formal bridge. Removing it would convert the theory into a component decomposition and conceal the proposed explanatory reversal.

4.3 Intervention and causal explanation

Interventionist accounts characterize causal relations through changes under suitable manipulations and connect explanation to invariance [18, 19]. SIM inherits that counterfactual discipline but does not infer constitution from intervention sensitivity alone. Many causally relevant variables can be changed without constituting the identity of a whole. The added requirements are generation relative to an isolated reference, non-reduction, specific return, registered empirical minimality, and representation covariance. “Constitutive” therefore names a constrained pattern of counterfactual evidence, not a synonym for “causally influential.”

4.4 Relational physics and standard decomposition

Relative coordinates and reduced mass are standard two-body mechanics, and relational interpretations of physical state long predate this work [20, 21]. Likewise, quantum chemistry contains mature families of energy, orbital, fragment, and atoms-in-molecules decompositions [22, 23, 24, 25]. SIM claims neither the discovery of the reduced coordinate nor decomposition itself.

Its atomic claim is methodological and interpretive: a standard relational structure is preregistered as a candidate closure, and limiting transformations and holdout observables test whether it organizes whole-formation readouts. Its molecular claim is a standard-theory constitutive counterfactual: the complete molecule-minus-isolated difference is removed, partially deleted, restored, and tested under representation controls. These operations ask whether the registered relation constitutes the tested molecular whole; they do not provide a new force or an exclusive route to the underlying numerical values.

4.5 Autopoiesis and organizational closure

Autopoiesis defines living organization through a network that produces and realizes the unity of its own components [26, 27]. Organizational and constraint-closure approaches further analyze mutually dependent constraints, self-maintenance, autonomy, and biological function [28, 29, 30, 31]. E009 is not presented as the first closure account of life and does not establish autopoiesis in a living cell.

The difference is experimental form. E009 does not begin by labeling a completed cell autopoietic. It recovers a distributed third-perspective candidate from differentiated boundary, metabolism, and information-repair dynamics, then subjects that candidate to selective erasure, temporal displacement, timely reinjection, severe-damage, and nonliving controls. The result is a reduced-model confirmation of a specified intervention–re-entry criterion for continued-whole formation. It is compatible with organizational closure but neither identical to it nor a replacement for its broader biological requirements.

4.6 Boundary of the SIM contribution

Table 2 makes the novelty claim explicit. The mathematical ingredients have precedents; the proposed contribution is their organization around perspective intersection and prospectively testable constitution.

Table 2: Neighboring concepts and the additional SIM commitment.

Neighboring work	Established contribution	Additional SIM commitment
Process and relational accounts	Objects may be relationally or dynamically individuated	Register the intersecting perspectives and test generation, destruction, and return of the constitutive relation
Scientific representation and invariance	Models are selective and identity may survive coordinate change	Preserve the intervention–re-entry identity of C_D across preregistered representations
Interventionist causation	Causal claims can be tested by suitable manipulations	Require specific restoration, registered empirical minimality, and whole-formation criteria in addition to causal effect
Two-body and quantum-chemical decomposition	Relative coordinates and interaction decompositions are standard	Treat a frozen standard structure as a constitutive counterfactual rather than claim a new calculation
Autopoiesis and organizational closure	Living unity can depend on self-producing or mutually maintaining organization	Extract a candidate third perspective and test erasure, timing, reinjection, and nonliving nulls

Implementation-specific neighboring concepts. The originating recurrent-perspective experiments were implemented with two computational agents and must also be separated from familiar agent-model constructs.

4.7 A's model of B and B's model of A

A representation held by A about B belongs locally to A; the converse belongs locally to B. The present claim begins only after both directed traces have been isolated. Experiment 006A asks whether those traces remain additive. The disconnected dual relay demonstrates the additive case: both computational agents solve reciprocal recall, yet Equation (7) cancels to numerical zero. O3 denotes the additional jointly nonseparable organization found in interacting systems.

4.8 Shared memory and an installed third computational agent

A shared memory localizes common information in an explicitly accessible place. A third computational agent possesses its own independently installed state and update. Neither is present in Experiments 005–006A. The emergent O3 is reconstructed from distributed changes inside A and B after learning.

4.9 Theory of mind, communication, and relational models

Theory-of-mind models normally infer beliefs or goals attributed to another agent. Emergent-communication models learn messages or protocols. Graph and relational models learn edges or relation types. These are adjacent but distinct objects. O3 is defined by reciprocal formation, nonseparability, pair specificity, bilateral causal action, and self-re-entry, not by semantic attribution, message identity, or edge prediction alone.

4.10 Functional perspective and phenomenal subjectivity

The term *perspective* is used here in a mathematical-functional sense fixed by Equation (2). The experiment establishes that the extracted C satisfies the registered structural and causal conditions used to identify an operational O3 in this model class. Phenomenal experience and independent subjecthood require additional bridge criteria and are not variables in the present experiment.

4.11 Connection to Subjectivity Intersection Ontology

The companion ontology preprint, *Subjectivity Intersection Ontology: A Comparative Ontological Framework for Third Subjectivity* [33], defines subjectivity as perspective and distinguishes two related structures. Its ontological prototype is a simultaneous triadic closure of Absolute Subjectivity, Relative Subjectivity, and Intersection Subjectivity. For the intersection of multiple Relative Subjectivities, it separately gives the operational form $AB \to C \to AB$: differentiated perspectives generate an irreducible third perspective, and that third acts back upon the perspectives through which it arose.

The present program connects directly to the second structure. Its O_A and O_B are finite operational perspectives rather than claims about Absolute or Relative Subjectivity as such. Their reciprocal intersection generates a jointly nonseparable C; Experiment 006A distinguishes this C from two additive directed memories, and Experiments 006 and 006R test whether present C constrains later A, B, and C formation. The computational sequence

$$(O_A, O_B) \xrightarrow{\text{intersection}} O_3 \xrightarrow{\text{self-re-entry}} (O'_A, O'_B, O'_3)$$

therefore supplies an executable realization of the ontology's multiple-relative-subject operational grammar. The points of connection are differentiated perspectives, generation through intersection, irreducibility of the third term, bilateral return, and closure through renewed intersection.

The ontological prototype and the computational realization nevertheless remain different in kind. The ontology begins from primordial unity, treats subjectivity as ontological perspective, makes the constitution of its triad simultaneous, and proposes recurrence across levels. The experiment begins from two separately initialized finite systems, represents their histories numerically, observes a temporal sequence of state updates, and tests a counterfactually extracted contribution. It does not test primordial unity, the projection of Absolute into Relative Subjectivity, phenomenal subjecthood, or cross-level recurrence. Nor does temporal self-re-entry by itself establish the ontology's simultaneous constitutive closure.

The connection is therefore neither an identity claim nor a merely decorative analogy. It is a directional bridge. The ontology specifies the structural hypothesis and distinguishes its ontological prototype from its operational expression; the mathematics turns the operational expression into conditions that can succeed or fail. Conversely, the experiments constrain which computational organizations can legitimately model $AB \to C \to AB$. Within that bridge, the present C is an *operational third perspective*; its identification with Intersection Subjectivity remains a further bridge hypothesis requiring criteria that connect functional perspective to subjectivity itself.

5 Operational foundation: staged recurrent-perspective experiments

5.1 Design logic

The originating recurrent-perspective program separates six questions:

1. Can an explicitly constructed third return state satisfy operational O3 criteria?
2. Does it preserve directed difference, retained history, and pair specificity?
3. Can an O3-like contribution arise from ordinary reciprocal learning without an O3 state or objective?
4. Does the emergent contribution alter both participants and re-enter later intersections?
5. Is functional return distinct from an external analyst's ability to isolate the contribution?
6. Is distributed C one jointly nonseparable perspective rather than two additive cross-memories?

Confirmatory thresholds, seeds, controls, and decision rules were frozen before execution. Failed seeds were neither removed nor replaced.

5.2 Experiments 003R and 004: sufficiency and retained difference

Experiment 003R corrected an invalid earlier execution and tested a completed explicit C under the full registered control inventory. Among 128 confirmatory pairs, 126 met the registered Phase A Class 2, Phase B Class 2, and Phase B O3 criteria. Across 1,024 Class 0/Class 1 control evaluations, no false Class 2 declaration occurred. A matched generic recurrent control also passed in 126 of 128 pairs; the result supports realizability and transfer, not a unique implementation.

Experiment 004 held $A + B$ constant while reversing $A - B$, matched current inputs while varying earlier interaction history, erased return paths, and exchanged completed carriers between pairs. All four registered hypotheses passed in 128 of 128 pairs. A role-aware recurrent control also passed, identifying a broader architecture class.

5.3 Experiment 005: spontaneous intersection under ordinary learning

Two receivers with independent persistent states learned delayed reciprocal recall. After external symbols disappeared, A reported B's earlier symbol and B reported A's. The only training loss was the sum of the two ordinary cross-entropies. There was no C or O3 state, third computational agent, pair identifier, carrier label, carrier target, or specialized regularizer.

Four equal-capacity interacting architectures and an independent control were evaluated. Post-learning directed intersection contributions were deleted and compared with 64 receiver-norm-matched random directions. A run passed the registered top-0.95 condition when deletion of the extracted contribution exceeded at least the 95th percentile of those controls.

5.4 Experiments 006 and 006R: self-re-entry

Experiment 006 froze three consecutive signal-free transitions. At each transition it removed the current contribution, applied one unchanged recurrent update, measured the missing next-state contribution, and reconstructed it. All 96 interaction units transported the contribution above matched random directions at all three transitions. The complete decision nevertheless remained NOT SUPPORTED because the distributed architecture passed a matched individual-history gate in 17 of 24 seeds against the preregistered floor of 18.

Experiment 006R retained that failure and prospectively separated two claims. The primary conjunction tested functional return, bilateral action, cross-pair loss, independent controls, and reconstruction. A separate prediction tested architecture-dependent external identifiability using 48 new seeds per architecture.

5.5 Experiment 006A: one O3 or two memories

Experiment 006A tested the decisive ambiguity. Its disconnected dual relay contained two 12-dimensional blocks and exactly the same 486 active parameters, data, optimization budget, and reciprocal-recall objective as each interacting architecture. It could learn both directed recalls but contained no cross-coupling capable of producing a nonadditive joint term.

Four interacting architectures and the disconnected control were evaluated on 48 preregistered seeds, 4,096 held-out episodes, and three signal-free transitions. The ten primary conditions covered capacity, competence, joint norm, bilateral support, learning amplification, transport, bilateral output response, cross-pair exchange, and exact reconstruction. Failure of any condition prevented the primary supported decision.

6 Results

Table 3: Preregistered evidence from constructed return to emergent operational O3.

Study	Question	Main observation	Decision
003R	Can an explicit return structure satisfy operational O3 criteria?	126/128 pairs; 0/1,024 false Class 2 null calls	Sufficiency supported
004	Does it preserve directed difference, history, transport, and pair specificity?	Four hypotheses passed in 128/128 pairs	Supported
005	Can an intersection-specific contribution emerge without an O3 objective?	84/96 top-0.95; independent-control norm 0	Supported
006	Does present C re-enter later states under the original conjunction?	Transport 96/96; distributed separation 17/24 versus floor 18	NOT SUPPORTED
006R	Can self-re-entry and external identifiability be separated?	13/13 primary and 3/3 boundary checks passed	Supported
006A	Is distributed C one nonseparable O3 rather than two directed memories?	10/10 checks; interacting C nonzero; competent disconnected C at numerical zero	SUPPORTED

6.1 Spontaneous formation

All four Experiment 005 interaction architectures met competence criteria. The extracted directed component exceeded the top-0.95 deletion threshold in 84 of 96 units: 23/24 distributed, 22/24 central shared, 18/24 directional relay, and 21/24 crossbar. Deletion affected both receivers; the independent control produced no corresponding component.

6.2 Self-re-entry and identifiability

In Experiment 006, transport passed in all 96 units and exact reconstruction error was below 2.71×10^{-16}, but the original conjunction failed as registered. In Experiment 006R, all 13 revised primary conditions passed on unused seeds. All-three-transition transport occurred in 47/48 distributed, 48/48 central-shared, 48/48 directional-relay, and 47/48 crossbar units: 190/192 pooled units against a floor of 168.

Median transported fractions were 0.7941, 0.9336, 0.9207, and 0.9419; median alignments were 0.6487, 0.9039, 0.9209, and 0.9253. Cross-pair exchange increased loss and deletion changed both receivers in every architecture. External-identifiability counts were 29/48, 46/48, 47/48, and 41/48 respectively, supporting all three preregistered architecture-boundary conditions. O3 can therefore be functionally active even when its distributed realization is difficult for an external observer to separate from individual history.

6.3 006A confirmation of one distributed O3

Experiment 006A passed all ten primary conditions. All five architectures were competent in 48 of 48 seeds; minimum held-out both-correct accuracy was 0.9751. Median joint-term norms were 0.1555, 0.1303, 0.0936, and 0.1019 for the interacting architectures. The equally competent disconnected dual relay had median norm 2.24×10^{-17} and maximum norm 2.93×10^{-17}.

Median bilateral support was 1.0 in every interacting architecture and 0 in the disconnected control. Median transported fractions across three signal-free transitions were 0.5243, 0.5487, 0.4916, and 0.4863, compared with 0.000024 in the disconnected control. Removal changed both receivers' output probabilities in architecture-level medians of 1.0000, 0.9984, 0.9960, and 0.9979 of held-out episodes. Cross-pair substitution imposed positive median cross-entropy cost in all four interacting architectures. Maximum one-step reconstruction error was 5.69×10^{-16}.

The 720 transition rows comprise 48 seeds, five architectures, and three transitions. No seed was replaced, no threshold changed, and no NaN or Infinity occurred. Secondary random-direction rankings were not uniformly high; the 006A claim is joint nonseparability against the competent additive control, while Experiments 005 and 006R retain the separately registered random-direction evidence for the full directed carrier.

7 Atomic closure: finite-mass relation and a mixed empirical record

7.1 Atomic bridge

Experiment 008 transfers Equation (14) to a two-body charged bound system without introducing a third particle. For constituent positions r_1, r_2 and masses m_1, m_2, the center-of-mass coordinate is removed and the relational coordinate

$$r = r_1 - r_2 \tag{24}$$

with reduced mass $\mu = m_1 m_2/(m_1 + m_2)$ carries the registered atomic closure. A change in r updates both constituent coordinates with nonzero mass-weighted fractions. The proposed atomic whole is therefore located operationally in a finite-mass relational bound closure, not in either constituent alone.

The preregistered structural chain audited boundary-independent identity, additive composition coordinates, inverse-square spherical distribution, cutoff-free stable binding in three dimensions, and composition-compatible continuous norm-preserving re-entry. The empirical holdout was the leading finite-mass $1S$–$2S$ prediction

$$\nu_{1S-2S} = \frac{3}{4}\frac{R_\infty c}{1 + m_e/m_{\text{partner}}}, \tag{25}$$

evaluated on muonium and positronium, neither of which was used in the disclosed private development set.

Neither r nor μ is a SIM discovery. Their status as standard mechanics is essential to the test, because the claim concerns explanatory use rather than algebraic novelty. SIM supplies a constitutive criterion under which the relational coordinate is treated not merely as a convenient separation of center-of-mass motion, but as a registered closure whose limiting transformation changes specified whole-formation observables. The novelty claim is therefore the preregistered constitutive interpretation and bridge, not renaming reduced mass as O3 and not deriving a correction beyond standard physics.

7.2 E008 transfer result

All five structural gates and both empirical holdouts passed. The muonium relative error was 9.4120×10^{-6} and the positronium relative error was 6.7708×10^{-5}. Relative to the infinite-partner-mass control, the registered finite-mass prediction improved absolute error by factors of 512.84 and 14771.39, respectively. This supports the registered leading gross-structure transfer. It does not address higher-order QED, spin, annihilation, radiative, or nuclear-size corrections.

7.3 The complete atomic record

The atomic program did not end with the positive E008 conjunction. Table 4 retains the complete public outcome sequence.

Table 4: Complete public atomic result record.

Study	Frozen question	Main observation	Decision
008	Does the atomic-closure chain transfer to two unused two-body systems?	Five structural and two empirical gates passed	SUPPORTED ATOMIC CLOSURE TRANSFER
008A	Does the parameter-free bilateral-response postulate $\lambda = 1$ improve the D/H hyperfine ratio?	SI log error 0.001258; standard-contact log error 0.000171	STANDARD CONTACT SUPPORTED
008B	Does the frozen nuclear-g and representation factorization predict the Li-6/Li-7 interval ratio?	Observed 0.284012567; predicted 0.283993247; log error 6.80×10^{-5}	FULL FACTORISED GENERATOR SUPPORTED
008C	Does the lithium ratio transfer from $2S_{1/2}$ to $2P_{1/2}$?	Central value remained compatible, but the registered precision gate failed	INSUFFICIENT PRECISION
008D	Does a frozen mixed-rank generator predict the K-39/K-41 correction ratio?	Target-free ratio 1.4239826743 publicly frozen; no result execution is claimed here	REGISTERED OPEN TARGET
008E	Can the signed rank-separated potassium relation predict the complete K-40 correction?	0.0081070823 kHz frozen and DOI-timestamped; no eligible independent benchmark found	OPEN NOVEL PREDICTION

The rejected 008A postulate is not repaired by the later positive lithium result, and the 008C precision-limited outcome is not counted as confirmation. E008D remains a registered open target and E008E remains a prediction, not an observed success. This mixed record is part of the method: the constitutive standpoint generates hypotheses, but every numerical specialization remains answerable to its own control and measurement gate.

8 Molecular closure: the complete carrier as a constitutive counterfactual

8.1 From atomic subspaces to a molecular whole

For a molecular geometry R, Experiment 010 defines the complete molecular carrier in a matched finite basis as

$$C_M(R) = H_{\text{molecule}}(R) - H_{\text{isolated centres}}. \tag{26}$$

The isolated reference preserves four physical atomic basis subspaces while removing molecule-induced sectors. C_M is neither a new force nor a single bond integral. It is the complete difference that must be restored for those subspaces to form the registered molecular whole.

Private MOL-001–MOL-010 exploration fixed the decomposition and intervention logic on two- and three-centre systems. E010 prospectively transferred them to an unexecuted rectangular H_4^+ target with four hydrogen centres, charge $+1$, three electrons, and PySCF `spin=1` (doublet, $2S = 1$). Full configuration interaction energies and one-particle density matrices were evaluated across a two-dimensional geometry grid in STO-3G, 6-31G, and cc-pVDZ.

8.2 Complete and partial deletion

The registration required nine gates in every basis profile. Complete carrier deletion tested whether the molecular energy landscape disappeared while the four local subspaces remained. One-electron cross-centre deletion tested how much binding was lost. Deletion of one physical edge tested whether a local relational intervention changed the global optimum and all four centre populations. Exact sector reconstruction, four orbital-representation controls, and Shapley attribution audited coordinate dependence and residual sectors.

Table 5: Experiment 010 registered four-centre result.

Basis	a/b at minimum (Å)	6 Å landscape depth (Eh)	Removed	Edge displacement (Å)	Gates
STO-3G	0.90/1.60	0.343822061	101.041%	1.221	9/9
6-31G	1.70/0.90	0.267435658	106.075%	0.600	9/9
cc-pVDZ	1.50/0.90	0.307894926	107.767%	1.000	9/9

The reported depth is the registered landscape depth relative to the frozen 6 Å reference, not a general dissociation claim. Because the rectangular H_4^+ landscape has symmetry-equivalent minima under $a \leftrightarrow b$, edge-deletion displacement is measured to the nearest member of the complete-system equivalent-minimum set; the registered threshold still passes under that convention.

The decision was COMPLETE MOLECULAR CARRIER TRANSFER SUPPORTED. The constitutive conclusion is counterfactual: the matched isolated centres do not retain the registered molecular landscape without C_M, while changing one edge reorganizes geometry and every local population. The calculation uses standard nonrelativistic finite-basis quantum chemistry. It does not show that Hamiltonian sectors are unknown to quantum chemistry, that Subjectivity-Intersection Mathematics uniquely predicts the numbers, or that an ontological O3 has been independently measured.

Quantum-chemical decomposition itself is not claimed as novel. The SIM contribution is to preregister one complete decomposition as a constitutive counterfactual and ask whether removal, incomplete return, and exact restoration determine the registered whole-level energy, geometry, and population readouts. The explanatory order, not the availability of the Hamiltonian terms, is the proposed innovation.

9 Cellular closure: natural emergence and reduced-model continued-whole formation

9.1 Reversing the object-first order

Experiment 009 begins with differentiated boundary, metabolism, and information-repair processes rather than a completed cell carrying those functions. No state named O3, life, viability, fitness, or global closure is installed. Reciprocal local laws generate trajectories from which a lagged cross-influence matrix is recovered. Its leading distributed mode is the registered cellular third-perspective candidate.

The explanatory question is constitutive rather than classificatory: does that relation merely correlate with recovery inside an already given cell, or does its correctly timed return satisfy the registered criterion for continued-whole formation? The registration therefore couples emergence and predictive self-re-entry to destructive, temporal, restorative, severe-damage, and nonliving controls. Terms such as "dynamic life constitution" are shorthand only for this reduced-model criterion; they are not a general definition or laboratory demonstration of life.

9.2 Emergent mode and self-re-entry

The recovered leading mode loaded on all three processes: 0.35057 on boundary, 0.32916 on metabolism, and 0.87679 on information-repair. Its participation ratio was 1.61858, above the registered 1.50 floor. Adding the other modules to each module's self-only next-change predictor improved held-out R^2 by 0.18593, 0.95895, and 0.76423. The minimum native gain exceeded twice the larger relation-erasure or time-shift control, and coordinate rescaling changed the gains by at most 2.22×10^{-16}.

Table 6: Experiment 009 held-out closure interventions, 32 lineages per condition.

Condition	Survived	Total	Constitutive readout
Native moderate damage	32	32	Recovery with median one division
Selective joint erasure	0	32	Local components retained; generated relation removed
Ten-minute relation time shift	0	32	Relation retained with incorrect temporal order
Timely relation reinjection	32	32	Recovery restored without adding an O3 substance
Native severe damage	0	32	Beyond the registered self-remaking boundary
Matched nonliving control	0	32	Activity without registered living closure
Undamaged native control	32	32	Baseline continuation

All fourteen gates passed, yielding REDUCED-MODEL DYNAMIC O3 SELF-REENTRY CLOSURE SUPPORTED. Within this realization, the modeled cell is not treated as the prior container of the three processes. Its registered continued-whole state is the higher-order dynamic closure constituted when their differentiated operational perspectives generate a third perspective whose self-re-entry remakes both the processes and the conditions of continued intersection.

The local laws and thresholds were result-informed by private development and the public result concerns new seeds in frozen reduced stochastic dynamics anchored to JCVI-syn3A source files. It does not establish the same O3 in the complete hybrid CME–ODE model or in living cells.

10 Within-bridge precursor: covariance of closure identity in E011

E011 motivated the later cross-domain sequence but did not itself execute independent atomic, molecular, and cellular targets. It is classified here as a prospective held-out consistency and representation-covariance test inside one frozen reduced cellular bridge. This reclassification leaves its preregistered decision and numerical results unchanged while specifying the evidential work that the result can perform.

10.1 From a repeated number to a preserved relation

Exploratory integration initially asked whether atomic, molecular, and cellular third terms share a raw numerical threshold. The apparent value near 0.72 did not survive changes of representation. The stronger invariant candidate was a transformed mediator relation. If λ is a lower-scale relational coupling, the frozen bridge defines lower-scale integrity λ^2 and a representation exponent q by

$$m = \lambda^{2q}. \tag{27}$$

Exploration at $q = 0.5, 1, 2$ fixed the median 90% mediator threshold at $m_{90} = 0.506944$. E011 then predicted raw thresholds at three unexecuted representations by

$$\lambda_{90}(q) = m_{90}^{1/(2q)}. \tag{28}$$

10.2 Held-out prediction

Each representation used 96 new stochastic lineages at every coupling value. All curves and all seven gates were frozen before the first target execution.

Table 7: Experiment 011 held-out within-bridge representation predictions.

q	Predicted λ_{90}	Observed λ_{90}	Absolute error	Observed m_{90}
0.75	0.635779702	0.634	0.001779702	0.504816901
1.25	0.762050926	0.762	0.000050926	0.506859310
1.50	0.797357951	0.798	0.000642049	0.508169592

All seven gates passed. The raw threshold range was 0.164, deliberately exceeding the registered 0.100 non-invariance floor. The transformed mediator range was 0.00335269, below the 0.020 ceiling, and transformed survival curves had pooled RMSE 0.00419620 against a 0.050 ceiling. The result supports *representation-covariant closure inside the frozen reduced bridge*: the surface coordinate moves, while the effective relation predicts the transformed threshold.

Equation (27) is an operational bridge law, not a derived microscopic law of nature. E011 does not show that natural atoms, molecules, and cells are physically connected by this numerical equation. Its contribution is mathematical and prospective: identity of a third term need not mean equality of raw coordinates; it can be defined by preserved causal or threshold structure under a registered representation map.

Because m is the only way λ and q enter the reduced dynamics, agreement in m-space primarily tests numerical and stochastic stability of the encoded bridge. Atomic and molecular results supplied the research motivation and history, not executed E011 targets. This limitation led directly to experiments in which no common scalar entered the three engines and the bridge law itself could fail.

11 Failure-guided independent cross-domain tests

11.1 Experiment 012: uniform intervention transfer is not supported

E012 was the first preregistered experiment in this sequence to execute independent reduced atomic, molecular, and cellular engines under the same four-condition logic: intact relation, O3 removal, correct O3 return, and mismatched return. No shared scalar mediator entered the domain dynamics. Atomic transfer passed all gates for hydrogen and deuterium; cellular transfer passed all gates in four previously unused seed cohorts; and the cc-pVDZ molecular realization passed all gates. The STO-3G and 6-31G molecular realizations, however, failed the registered mismatched-return gate, and STO-3G also failed the specific-return advantage.

The preregistered overall decision was therefore

NOT SUPPORTED: CROSS-DOMAIN O3 INTERVENTION TRANSFER.

This was not a null result across every component. Removal reduced the registered score in every realization, correct return restored a high score in every realization, and the threshold-free separation margin remained positive at 0.207458764. What failed was the stronger claim that the complete four-condition signature, including mismatch specificity under common decision gates, transfers uniformly across every tested representation. E012 localized the failure to representation-sensitive molecular mismatch discrimination.

11.2 Experiment 013: domain-prior generation without common calibration

E013 asked whether O3 candidates and mismatches could be generated from domain-prior rules rather than imported outcome values. The residual rule and exact reconstruction passed in all three domains, and structurally generated mismatches passed the registered centered-overlap bound. Both fresh atomic targets and all four fresh cellular cohorts passed the absolute gates. The fresh molecular realization also satisfied all four strict causal inequalities:

$$S_D(X_D^0) > S_D(X_D^-), \quad S_D(X_D^0) > S_D(X_D^\times), \quad S_D(X_D^+) > S_D(X_D^-), \quad S_D(X_D^+) > S_D(X_D^\times). \tag{29}$$

Nevertheless, its removal score was 0.70875 and mismatched-return score was 0.80625, both above the preregistered common maximum of 0.50. The overall decision was therefore

NOT SUPPORTED: DOMAIN-PRIOR O3 GENERATION TRANSFER.

The threshold-free separation margin was positive at 0.161222491. E013 supports the component findings of domain-prior residual generation, exact reconstruction, and ordinal causal separation in the executed reduced models. It does not confirm transfer of common absolute calibration. Its failure identifies a sharper candidate invariant: the causal ordering of correct return relative to removal and mismatch, expressed through domain-native readouts rather than a universal score threshold.

12 Prospective stabilized O3-return specificity: E014

12.1 Registered hypothesis and design

E014 froze the refinement produced by E012 and E013. For each domain it preregistered a fresh target, a domain-native stabilization interval, an intact reference, removal, correct return, a physically distinguishable mismatch selected without intervention outcomes, and the paired margins in Equation (21). The three engines shared no scalar mediator. The cross-domain claim would fail if either margin were non-positive at any registered primary checkpoint in any domain.

The target engines were hydrogenic helium-4 in an atomic soft-Coulomb relaxation, a fresh H_4^+ cc-pVTZ line at $b = 1.1$ Å in the molecular engine, and two held-out damage amplitudes in the frozen E009 reduced stochastic cell. The readouts were intentionally domain-specific: Hellinger similarity for the atomic state, registered molecular landscape similarity across grid refinements, and paired multivariate state similarity plus the preregistered recovery boundary for the cellular system.

12.2 Prospective result

All structural gates and all three domain decisions passed. Atomic helium-4 passed $3/3$ stabilized checkpoints; the minimum correct-return advantage was 0.258272030 over removal and 0.285485351 over mismatch. The molecular target passed $9/9$ checkpoints across grid factors 1, 2, and 4; the corresponding minimum advantages were 0.714143349 and 0.877924653. In the cell model, reinjection at 45 minutes yielded alive fraction 1.0 in every cohort at both held-out damage amplitudes, while the registered 50-minute delayed return yielded 0.0. All $8/8$ stabilized cellular checkpoints passed; across the 16 paired comparisons, the minimum mean margin was 0.039248630 and the minimum one-sided 99% bootstrap lower bound was 0.037755892.

The preregistered overall decision was

SUPPORTED: STABILIZED O3 RETURN CROSS-DOMAIN TRANSFER.

Thus, over the tested realizations, the shared object is the direction of stabilized causal specificity:

> After a domain-native stabilization interval, correct O3 return is closer to the registered domain state than continued O3 removal or a physically distinguishable mismatched O3 return.

Three prospective domain-specific tests passed. Leave-one-domain-out recombinations restated the same direction but are not additional independent transfers: they recombine already evaluated domain-pass values rather than learn a source-level mechanism from two domains and apply it to the third.

12.3 What E014 adds and what it does not

E014 is stronger than E011 in one precise respect: atom, molecule, and cell are executed as independent targets, and success is not an algebraic consequence of one mediator inserted into all engines. It is narrower than an assertion of source identity. Equation (22) preserves an ordering after registered stabilization; it does not equate the atomic carrier, molecular Hamiltonian difference, cellular mode, metrics, timescales, or mechanisms. The cellular persistence checkpoints are temporal evidence inside the reduced model, not evidence for a long-range spatial or nuclear analogue.

The combined E012–E014 record therefore does not obtain a positive result by relaxing an unchanged hypothesis after failure. Each registered failure removes a stronger candidate: representation-independent mismatch thresholds in E012 and common absolute calibration in E013. E014 prospectively risks the surviving hypothesis—domain-native stabilized causal direction—on fresh targets and checkpoints.

13 Cross-domain synthesis

13.1 One grammar, domain-specific third terms

Table 8 states the synthesis without collapsing the domains into one substance or one estimator.

Table 8: Domain realizations of the constitutive-closure grammar.

Domain	Differentiated operational perspectives	Generated third term	Selective counterfactual	Whole-formation readout
Recurrent-perspective system	Two recurrent history organizations	Inclusion–exclusion joint contribution	Erasure, exchange, disconnected additive control	Bilateral next-state and next-intersection change
Atom	Finite-mass charged constituents	Relative-coordinate bound closure	Infinite-mass and registered structural controls	Leading finite-mass spectrum and bilateral coordinate update
Molecule	Matched isolated atomic subspaces	Complete molecule-minus-isolated Hamiltonian carrier	Complete, sector, and single-edge deletion	Energy landscape, optimum geometry, all-centre populations
Cell	Boundary, metabolism, information-repair processes	Recovered distributed cross-influence mode	Joint erasure, time shift, reinjection, severe and nonliving controls	Recovery, division, and next-change prediction in every process
Within-bridge covariance (E011)	Alternative representations of one frozen reduced bridge	Effective transformed mediator relation	Held-out representation exponents	Predicted raw thresholds and transformed curve collapse
Independent cross-domain sequence (E012–E014)	Domain-relative atomic, molecular, and cellular perspectives	Domain-specific O3 candidate under removal and return	Correct return versus removal and physical mismatch after stabilization	Sign of paired domain-native similarity margins

The shared achievement is not a common raw number. It is an experimentally disciplined reversal of explanatory order. Instead of beginning with a completed object and measuring relations inside it, each transfer begins with differentiated operational perspectives and asks whether their joint relation is necessary for the registered formation readout. The recurrent-perspective result identifies an operational O3; the physical and biological transfers then ask whether domain-specific third perspectives play an analogous constitutive role under independently fixed bridges. E014 adds that the direction of correct O3 return, after domain-native stabilization, survives direct prospective execution in all three domain engines.

13.2 What the synthesis does not identify

The distributed neural contribution, atomic relative-coordinate carrier, molecular Hamiltonian difference, cellular mode, and transformed mediator are not asserted to be mathematically identical objects. They occupy different state spaces, use different interventions, and carry different empirical status. E012 and E013 further show that neither a common absolute score nor uniformly calibrated mismatch suppression is warranted. The cross-domain result concerns a shared operational grammar and a prospectively preserved causal direction. An ontological claim that all domain third terms are one substance or instances of one physically identical mechanism remains a further bridge hypothesis.

14 Discussion

14.1 Why C is a third perspective

C is not called O3 merely because it is a third mathematical term. Under the operational definition, a perspective is a history-bearing organization that transforms received information and constrains subsequent formation. C satisfies that role jointly: it is generated only by reciprocal intersection, has support across both receivers, loses function when either distributed realization is treated separately, changes both participants when removed, is pair-specific, and contributes to the next participant states and next intersection.

The decisive contrast is between competence and intersection. The disconnected relay remembers both directions and solves the same task, but its two memories remain additive and produce numerical-zero joint C. The interacting systems reach comparable competence while producing a nonzero jointly active term. O3 is therefore not equivalent to A's memory of B plus B's memory of A. It is the non-separable organization of their reciprocal taking-up.

14.2 Nonlocalization

No central third object appears in the distributed architecture. One realization of O3 is present in how A has been changed through B; the other is present in how B has been changed through A. Neither contains the whole. Operational unity is established by their inseparable causal function, not by spatial co-location. Experiment 006A supplies the missing test that allows the two distributed realizations to be treated as one O3 rather than named together by convention.

14.3 Self-re-entry as constraint

Transport is stronger than persistence of a stored item. Removing C_t changes $h_A(t+1)$ and $h_B(t+1)$ under the same update rule, and changes the next extracted C_{t+1}. The present third perspective therefore constrains both the next differentiated perspectives and the next event of intersection. This realizes the computational form of $AB \to C \to AB$ as

$$(O_A, O_B) \to O_3 \to (O'_A, O'_B, O'_3).$$

14.4 What changed from relational generation

Relational generation remains an important consequence, but it is no longer the primary explanatory term. A relation can be externally described without either participant taking up the other. The experiments are more specifically about reciprocal perspective formation. The relation becomes causally generative because intersection produces O3 and O3 re-enters subsequent formation. The conceptual order is therefore perspective difference, reciprocal intersection, emergent O3, and relational self-re-entry.

14.5 The role of the preserved failure

Experiment 006 remains negative under its original conjunction. This matters because it prevented functional re-entry from being equated with clean external separability. Experiment 006R then confirmed the separated hypotheses prospectively. The sequence supports a central feature of distributed O3: an operation may be causally coherent across A and B even when an external analyst cannot isolate it uniformly as a dominant direction in each participant.

The same discipline extends across domains. Experiment 008A rejects its SI-specific response scale, 008C remains unresolved, and 008E remains open. E012 rejects uniform transfer of the complete four-condition signature, and E013 rejects common absolute calibration despite positive component results. These outcomes prevent the positive E008, E008B, E009, E010, E011, and E014 decisions from being converted into a general presumption of success. A cross-domain mathematics becomes scientific only when its bridges can fail independently and when later refinements are frozen before new targets are executed.

14.6 From detecting relations to redefining scientific objects

Existing scientific calculations need not be numerically wrong for their explanatory order to be reconsidered. E010 uses standard quantum chemistry and E008 uses standard finite-mass structure. The Subjectivity-Intersection contribution is to ask a prior constitutive question: which intersection must be present for differentiated operational perspectives to form the object whose properties are then calculated? In this order, an atom is not first assumed and then assigned a relative coordinate; a molecule is not first assumed and then assigned bonds; a cell is not first assumed and then assigned integrated functions. Each object is operationally located in a perspective intersection that generates and re-forms one whole under the relevant domain bridge.

This is a proposed redefinition of explanatory standpoint, not a claim that every conventional equation has been superseded. The existing formalisms supply indispensable coordinates, dynamics, measurements, and controls. Subjectivity-Intersection Mathematics reorganizes them around formation, selective destruction, and re-entry of the whole relation.

14.7 From representation covariance to causal-direction invariance

E011 adds a mathematical constraint to comparison within a frozen bridge. If the identity of closure were attached to a raw coordinate, it would be destroyed by the held-out representation changes. Instead, the raw threshold moved as registered while the transformed mediator threshold remained narrow. E012–E014 expose the stronger claim to independent domain engines. Their combined result relocates the transferable object again: not in a universal transformed number, but in the sign of the correct-return advantages after domain-native stabilization. This motivates an equivalence-class problem: determine which transformations preserve the generation, intervention history, and return ordering of C_D, even when its coordinates, localization, estimator, and timescale change.

15 Subjectivity-Intersection Mathematics as a research program

The program now has six connected layers:

1. **Perspective mathematics:** formalize differentiated, history-dependent state organizations.
2. **Intersection operations:** define reciprocal crossing, nonseparability, identity, exchange, and equivalence.
3. **Self-re-entry dynamics:** characterize how generated O3 constrains subsequent perspectives and intersections.
4. **Domain constitutive bridges:** specify local terms, isolated references, interventions, and whole-formation readouts for physical and biological targets.
5. **Representation covariance:** characterize which features of a third term remain invariant when raw coordinates and implementations change.
6. **Cross-domain causal invariance:** test whether a preregistered intervention–return direction survives independent domain engines without a common scalar mediator.

Candidate invariants across the tested implementations are preserved differentiation, non-reduction to isolated terms, joint causal necessity, return into the terms that constitute the system, intervention identity, reconstruction or covariance under registered representation changes, stabilized correct-return specificity, and the distinction between functional activity and external identifiability.

Priority mathematical problems include necessary and sufficient conditions for O3 emergence; equivalence classes of distributed, localized, physical, and dynamical third terms; conservation and transformation laws under repeated intersection; extension beyond two participants; continuous-time intersection; and representation theorems that distinguish genuine constitutive covariance from a transformation built into the model. Priority cross-domain problems are tests in which a causal signature is learned without outcome leakage from two independently generated domains and then applied to a third, together with designs that can distinguish equal readout direction from common source construction. Priority empirical problems are independent evaluation of the K-40 prediction, higher-precision lithium cross-state measurements, molecular transfer to non-hydrogenic and experimentally constrained targets, and cellular interventions whose observables can be measured in living or complete mechanistic systems.

16 Scope and limitations

The evidence is heterogeneous by design and must remain so in interpretation. The recurrent-perspective result concerns finite recurrent systems trained on synthetic reciprocal recall. The atomic result combines structural audits with leading published benchmarks and reduced dynamical realizations; it does not replace higher-order atomic theory. The molecular result is a computational counterfactual inside standard finite-basis quantum chemistry and reduced landscape dynamics. The cell result concerns result-informed reduced stochastic dynamics on new registered seeds. E011 concerns a frozen power-law bridge and does not derive that bridge from microscopic physics. E012–E014 execute independent reduced domain engines, but their common result remains an operational sign invariant rather than a shared microscopic model.

The operational identification of C with O3 is a theoretical interpretation formulated after the preregistered 006A result. Experiment 006A preregistered joint nonseparability, not the wording of this subsequent identification. Future tests can preregister the complete O3 definition given in this paper.

The extraction operators are domain-specific. Equation (7), the atomic relative coordinate, Equation (26), the cellular lagged mode, and Equation (27) are not proved unique. Registered empirical minimality is therefore limited to the alternatives and interventions specified in advance. Other operators may reveal equivalent or different intersection structures. The current results establish reproducible operational objects and interventions, not a universal representation theorem or necessary-and-sufficient conditions across all possible decompositions.

The words *perspective* and *O3* are nevertheless retained deliberately. They do not attribute consciousness to atoms, molecules, cellular modules, or numerical modes. They denote the domain-relative distinctions, transformations, and causal readouts formalized in Equation (12). An operational third perspective is not by itself evidence of phenomenal experience or independent subjecthood. Identification of the domain third terms with the ontology's Intersection Subjectivity remains a bridge hypothesis.

The results do not establish a universal constant near 0.72, golden-mean or pentagonal geometry, a new force, an ab initio atom-to-cell causal chain, common O3 substance, or laboratory confirmation of living-cell O3. They do not establish representation-independent mismatch scores or common absolute calibration: those stronger registered claims failed in E012 and E013. The supported claim is narrower and more foundational: constitutive closure can be defined through differentiated perspectives, a generated joint relation, selective intervention, return, and representation controls; across the tested independent domains, the stabilized direction of correct O3 return can then be subjected to and survive a prospective decision.

17 Conclusion

The originating experiments established that two interacting recurrent perspectives can generate one distributed operational O3 absent from an equally competent additive pair, and that present O3 can constrain the next two states and next intersection. This paper asks what follows when this experimentally disciplined grammar is transferred to scientific objects whose wholeness is usually presupposed, while making explicit that an operational perspective is more than a local component and less than a claim of phenomenal consciousness.

The atomic sequence supports a finite-mass closure transfer while retaining a rejected postulate, a precision-limited result, and an open prediction. The molecular experiment supports the complete carrier's prospective four-centre transfer inside standard quantum chemistry. The cellular experiment supports natural emergence and timed self-re-entry of a distributed mode as the condition of reduced-model recovery. E011 supports a held-out representation-covariant threshold relation inside one frozen reduced bridge rather than a universal raw number. The negative E012 and E013 decisions then narrowed the transferable candidate from common calibration to domain-native causal direction, which

E014 tested prospectively.

E014 prospectively executed independent atomic, molecular, and cellular targets without one shared scalar mediator. Across all registered primary checkpoints, correct O3 return after domain-native stabilization was closer to the registered state than continued removal or physically distinguishable mismatched return. This supports a cross-domain causal invariant of stabilized O3-return specificity over the tested reduced realizations. It does not identify a common substance, common metric, or source-level mechanism.

The two-participant statement remains

$$\boxed{(O_A, O_B) \xrightarrow{\text{intersection}} O_3 \xrightarrow{\text{self-re-entry}} (O'_A, O'_B, O'_3)}$$

and its domain-relative generalization is

$$\boxed{(O_1^D, \ldots, O_n^D) \xrightarrow{\mathcal{X}_D} C_D \xrightarrow{\mathcal{E}_D} (O_1^{D\prime}, \ldots, O_n^{D\prime}, C'_D)}.$$

Subjectivity-Intersection Mathematics is therefore proposed as a mathematics of constitutive closure: how differentiated operational perspectives intersect, how a non-reducible third perspective is generated, how it returns into the formation of its perspectives, and how the identity of that closure can be tested across interventions and representations. Its cross-domain achievement is not to replace existing calculations wholesale or to equate domain carriers, but to show prospectively that the causal direction of stabilized O3 return can survive independently executed domain realizations after stronger scalar and calibration claims have failed. The explanatory order is thereby reversed from an already existing object with relations to an intersection whose generation and return are experimentally tested as conditions of the object's continued constitution.

Acknowledgements

The author gratefully acknowledges Marcel Wende, Luke Leighton and C.A.T., and Thomas Loker. Through their rich insights, ideas, and suggestions, they patiently sustained the collaborative research that provided the foundation from which this paper emerged. Marcel Wende, in particular, contributed with exceptional persistence through his consistent auditing of the research program, its experimental claims, and their evidential boundaries.

A Reproducibility and public records

Repository

`https://github.com/SIEL-Research/Subjectivity-Intersection-Mathematics`

Zenodo community

`https://zenodo.org/communities/subjectivity-intersection-mathematics`

Study	GitHub result implementation	Zenodo result and preregistration
003R	Corrected operational sufficiency experiment	Result: doi:10.5281/zenodo.21761140; preregistration: doi:10.5281/zenodo.21761030
004	Difference-preserving and retained-history audit	Result: doi:10.5281/zenodo.21762041; preregistration: doi:10.5281/zenodo.21761975

Study	GitHub result implementation	Zenodo result and preregistration
005	Emergent carrier solution class	Result: doi:10.5281/zenodo.21763963; preregistration: doi:10.5281/zenodo.21763898
006	Spontaneous operational O3 re-entry	Result: doi:10.5281/zenodo.21764472; preregistration: doi:10.5281/zenodo.21764327
006R	Revised confirmation and architecture boundary	Result: doi:10.5281/zenodo.21764580; preregistration: doi:10.5281/zenodo.21764531
006A	Emergent nonseparable relational C	Result: doi:10.5281/zenodo.21785748; preregistration: doi:10.5281/zenodo.21785602
008	Atomic-closure transfer	Result: doi:10.5281/zenodo.21853661; preregistration: doi:10.5281/zenodo.21853576
008A	Bilateral-response hyperfine test	Result: doi:10.5281/zenodo.21854058; preregistration: doi:10.5281/zenodo.21854011
008B	Lithium isotope representation prediction	Result: doi:10.5281/zenodo.21855898; preregistration: doi:10.5281/zenodo.21854314
008C	Lithium cross-state transfer	Result: doi:10.5281/zenodo.21854663; preregistration: doi:10.5281/zenodo.21854601
008D	Potassium mixed-rank registered target	Preregistration: doi:10.5281/zenodo.21854993; no result decision claimed
008E	K-40 signed rank-separated prediction	Result/search closure: doi:10.5281/zenodo.21855249; preregistration: doi:10.5281/zenodo.21855190
009	Constitutive cell O3 closure	Result: doi:10.5281/zenodo.21862060; preregistration: doi:10.5281/zenodo.21862026
010	Complete molecular-carrier transfer	Result: doi:10.5281/zenodo.21865750; preregistration: doi:10.5281/zenodo.21865563
011	Renormalized cross-scale closure prediction	Result: doi:10.5281/zenodo.21875050; preregistration: doi:10.5281/zenodo.21874997
012	Independent cross-domain O3 intervention transfer	Result: doi:10.5281/zenodo.21888156; preregistration: doi:10.5281/zenodo.21888131
013	Domain-prior O3 generation transfer	Result: doi:10.5281/zenodo.21894771; preregistration: doi:10.5281/zenodo.21894665
014	Stabilized O3-return cross-domain transfer	Result: doi:10.5281/zenodo.21895729; preregistration: doi:10.5281/zenodo.21895645

The invalid Experiment 003 record remains preserved at doi:10.5281/zenodo.21760837; the Experiment 003R technical addendum is doi:10.5281/zenodo.21761925. Experiment 006 remains NOT SUPPORTED. Experiment 008A retains its standard-contact decision, 008C remains precision-limited, 008D remains unexecuted, and 008E remains an open externally testable prediction. Experiments 012 and 013 retain their preregistered NOT SUPPORTED decisions; their positive component results are not promoted to overall confirmation. Experiment 014 retains its preregistered SUPPORTED decision.

B Terminology

Operational perspective
A history-dependent state organization that transforms received information and constrains subse-

quent state and output formation.

Domain-relative operational perspective
A tuple $O_i^D = (L_i, S_i, T_i, Q_i)$ comprising a local carrier, the distinctions available from its relational position, admissible transformations, and registered causal readouts. It is not synonymous with the carrier and does not imply consciousness.

Reciprocal perspective intersection
Mutual state formation in which A takes up B through A's organization and B takes up A through B's, while their distinction is preserved.

Directed intersection trace
One receiver-side change attributable to the other under a matched counterfactual.

Operational third perspective O3
A post-learning, jointly nonseparable, pair-specific and self-re-entering contribution generated by reciprocal perspective intersection and distributed across both participants.

Nonseparable C
The inclusion-exclusion term $H(ab) - H(a0) - H(0b) + H(00)$, which vanishes for additive disconnected memories.

Self-re-entry
Contribution of present O3 to the next participant states and next intersection through unchanged dynamics.

External identifiability
An analyst's ability to separate O3 from a matched individual-history contribution; it is distinct from O3's causal function.

Domain-relative local carrier
The physical, biological, or computational term L_i that realizes one operational perspective; by itself it does not exhaust the perspective.

Constitutive closure
A bridge-relative structure satisfying the applicable registered conditions of generation, necessity, re-entry, specificity, registered empirical minimality, representation covariance, and null discrimination.

Complete molecular carrier
The matched-reference Hamiltonian difference $H_{\text{molecule}} - H_{\text{isolated centres}}$ used in E010; it is not a new force.

Representation-covariant closure
Preservation of a registered closure relation or decision under a specified transformation even when the corresponding raw coordinates differ.

Stabilized O3-return specificity
The registered condition that, after a domain-native stabilization interval, correct O3 return is closer to the intact domain state than continued removal or a physically distinguishable mismatched return.

Cross-domain causal-direction invariant
Preservation of the sign vector $(\operatorname{sgn}\Delta_D^-, \operatorname{sgn}\Delta_D^\times) = (+,+)$ across independently executed domains; it does not assert equality of scores, carriers, substances, or source mechanisms.

Compression ladder
The one-way evidential sequence from source construction through O3 carrier, intervention history,

domain state, and registered readout to a compressed invariant. Agreement at a later level does not by itself identify an earlier-level object.

Evidence class
The explicit status of a result as benchmark-grounded transfer, standard-theory counterfactual, reduced-model confirmation, reduced-bridge prediction, or independent cross-domain intervention.

Selected references

[1] Battaglia, P. W., Hamrick, J. B., Bapst, V., et al. (2018). Relational inductive biases, deep learning, and graph networks. *arXiv:1806.01261*.

[2] Kipf, T. N., Fetaya, E., Wang, K.-C., Welling, M., and Zemel, R. (2018). Neural relational inference for interacting systems. *Proceedings of ICML*, 2688–2697.

[3] Lazaridou, A., Hermann, K. M., Tuyls, K., and Clark, S. (2018). Emergence of linguistic communication from referential games. *NeurIPS 31*.

[4] Locatello, F., Bauer, S., Lucic, M., et al. (2019). Challenging common assumptions in unsupervised disentangled representations. *Proceedings of ICML*, 4114–4124.

[5] Nosek, B. A., Ebersole, C. R., DeHaven, A. C., and Mellor, D. T. (2018). The preregistration revolution. *PNAS*, 115(11), 2600–2606.

[6] Rabinowitz, N. C., Perbet, F., Song, H. F., et al. (2018). Machine theory of mind. *Proceedings of ICML*, 4218–4227.

[7] Rumelhart, D. E., Hinton, G. E., and Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323, 533–536.

[8] Santoro, A., Raposo, D., Barrett, D. G. T., et al. (2017). A simple neural network module for relational reasoning. *NeurIPS 30*.

[9] Sukhbaatar, S., Szlam, A., and Fergus, R. (2016). Learning multiagent communication with back-propagation. *NeurIPS 29*.

[10] Simondon, G. (1992 [1958]). The genesis of the individual. In J. Crary and S. Kwinter (Eds.), *Incorporations*, 296–319. Zone Books.

[11] Dupré, J. (2012). *Processes of Life: Essays in the Philosophy of Biology*. Oxford University Press.

[12] Nicholson, D. J., and Dupré, J. (Eds.). (2018). *Everything Flows: Towards a Processual Philosophy of Biology*. Oxford University Press.

[13] Worrall, J. (1989). Structural realism: The best of both worlds? *Dialectica*, 43(1–2), 99–124. doi:10.1111/j.1746-8361.1989.tb00933.x.

[14] Ladyman, J. (1998). What is structural realism? *Studies in History and Philosophy of Science Part A*, 29(3), 409–424. doi:10.1016/S0039-3681(98)80129-5.

[15] Giere, R. N. (2006). *Scientific Perspectivism*. University of Chicago Press.

[16] van Fraassen, B. C. (2008). *Scientific Representation: Paradoxes of Perspective*. Oxford University Press.

[17] Frigg, R., and Nguyen, J. (2020). *Modelling Nature: An Opinionated Introduction to Scientific Representation*. Springer. doi:10.1007/978-3-030-45153-0.

[18] Pearl, J. (2000). *Causality: Models, Reasoning, and Inference*. Cambridge University Press.

[19] Woodward, J. (2003). *Making Things Happen: A Theory of Causal Explanation*. Oxford University Press. doi:10.1093/0195155270.001.0001.

[20] Landau, L. D., and Lifshitz, E. M. (1977). *Quantum Mechanics: Non-Relativistic Theory*, 3rd ed. Pergamon Press.

[21] Rovelli, C. (1996). Relational quantum mechanics. *International Journal of Theoretical Physics*, 35, 1637–1678. doi:10.1007/BF02302261.

[22] Morokuma, K. (1971). Molecular orbital studies of hydrogen bonds. III. C=O...H–O hydrogen bond in H_2CO...H_2O and H_2CO...$2H_2O$. *The Journal of Chemical Physics*, 55, 1236–1244. doi:10.1063/1.1676210.

[23] Kitaura, K., and Morokuma, K. (1976). A new energy decomposition scheme for molecular interactions within the Hartree–Fock approximation. *International Journal of Quantum Chemistry*, 10(2), 325–340. doi:10.1002/qua.560100211.

[24] Bickelhaupt, F. M., and Baerends, E. J. (2000). Kohn–Sham density functional theory: Predicting and understanding chemistry. *Reviews in Computational Chemistry*, 15, 1–86. doi:10.1002/9780470125922.ch1.

[25] Stone, A. J. (2013). *The Theory of Intermolecular Forces*, 2nd ed. Oxford University Press.

[26] Varela, F. G., Maturana, H. R., and Uribe, R. (1974). Autopoiesis: The organization of living systems, its characterization and a model. *BioSystems*, 5(4), 187–196. doi:10.1016/0303-2647(74)90031-8.

[27] Maturana, H. R., and Varela, F. J. (1980). *Autopoiesis and Cognition: The Realization of the Living*. D. Reidel.

[28] Mossio, M., Saborido, C., and Moreno, A. (2009). An organizational account of biological functions. *The British Journal for the Philosophy of Science*, 60(4), 813–841. doi:10.1093/bjps/axp036.

[29] Mossio, M., and Moreno, A. (2010). Organisational closure in biological organisms. *History and Philosophy of the Life Sciences*, 32(2–3), 269–288.

[30] Montévil, M., and Mossio, M. (2015). Biological organisation as closure of constraints. *Journal of Theoretical Biology*, 372, 179–191. doi:10.1016/j.jtbi.2015.02.029.

[31] Moreno, A., and Mossio, M. (2015). *Biological Autonomy: A Philosophical and Theoretical Enquiry*. Springer. doi:10.1007/978-94-017-9837-2.

[32] Watanabe, S. (2026). *Subjectivity-Intersection Mathematics: A Mathematical Research Program for Viewpoints, Relational Generation, and Simultaneous Closure*. Research white paper. `https://www.researchgate.net/publication/410728192`.

[33] Watanabe, S. (2026). *Subjectivity Intersection Ontology: A Comparative Ontological Framework for Third Subjectivity*. Version 1.0.0. Zenodo preprint. doi:10.5281/zenodo.21641878.